

Rushil Shah

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SKILLS

Languages: Golang, Python, Java, C++, Ruby, Typescript, Javascript, Elixir, HTML5, CSS

Technologies: RabbitMQ, Kafka, React, Django, Phoenix, ElasticMQ, REST, InfluxDB, AWS(SQS, SNS, GraphQL, gRPC

Databases: PostgreSQL, Elasticsearch, MongoDB, DynamoDB, Firebase, Cassandra, Redis

PROFESSIONAL EXPERIENCE

Tesla | Software Development Engineer

Feb 2024 - Present

- Designed **Autosweep** to clear out old data based on business requirements, saving **33%** of Tesla Advisors time via **Goroutines** and **Cron Jobs**, configurable via **SQL** query without production releases for future changes.
- Engineered real-time email notification feature by programming cron jobs to poll data at regular intervals, and asynchronously send the email payload to necessary **RabbitMQ** queues for message transfer.
- Accomplished a **40%** reduction in write errors, due to version conflict while writing documents in ElasticSearch by implementing retry mechanism with error handling, random jitters and async programming, using **Goroutines**
- Enabled storing **Database transactions** in request context to avoid creating multiple transactions for every database call using **Golang Context** package, improving the CPU consumption by 20% and drastically reducing DB load

Fetch | Backend Engineer

Nov 2023 - Feb 2024

- Scaling microservices that scan millions of user receipts, by implementing microservices and pipelines in **Golang** to check validations, item expiry, normalization, etc, and storing data in the **Snowflake** data lake
- Managed migration of **314 GB** data from **AWS ElastiCache** to a compact version and **DynamoDB**, optimizing CPU utilization from **95%** to **32%** through effective fallback and feature flag implementation
- Upgraded legacy endpoints to microservices, optimized data flow with **AWS Lambda** updates, and improved monitoring via **Grafana** and **Elastic Beanstalk**. Enhanced system agility, performance, and reliability

Influxdata | Software Engineering Intern ([General Marketplace Simulator](#))

May 2022 - Aug 2022

- Designed a **Notification service** to simulate the **AWS Marketplace** for testing the onboarding of **70%** of company users, using the functional programming in **Elixir Phoenix**, to send messages to local queues in **Docker**
- Conceptualized and implemented the architecture to duplicate **AWS SNS** and **SQS** services locally in **Elixir**, by sending notifications at run time and polling message queues at regular intervals, locally using **ElasticMQ**

Shipmnts | Full Stack Software Engineer

Dec 2020 - Jun 2021

- Improved scalability by **30%** in a Subscription service that delivers tracking data in real-time, by using a **queueing architecture** in **Ruby on Rails** to delegate large asynchronous tasks to different queues for information retrieval
- Achieved a **50%** reduction in API consumption in a subscription service by programming **Cron Jobs** in **JavaScript** to fetch and update data asynchronously, and adjusting cron time intervals at runtime based on previous tracking data

Taasha Technologies | Full Stack Software Engineer

Jun 2020 - Sep 2020

- Developed an E-commerce application for over **20000** products, using **Java Spring**, **React**, and **RESTful APIs** in a microservice architecture, which included services like catalog, user authentication, order processing, etc
- Led the optimization of search filters and latency reduction efforts, resulting in a **30%** improvement in performance using **PostgreSQL** and **Redis** by implementing a hierarchical data structure and writing efficient **SQL** queries

EDUCATION

Master of Science in Software Engineering | San Jose State University, California - **GPA: 3.81/4**

Aug 2023

PROJECTS

Picture-To-Product: Golang | Python | NodeJS | React Hooks | Javascript | EC2

Feb 2023 - May 2023

- Developed an image classification project using **Convolutional Neural Networks** (CNN) machine learning model in **Python** to classify products based on images accurately and exposed model services using **Python Django** API
- Constructed data pipelines and API integration in **Golang** and **AWS EC2** to seamlessly pass product data to the classification service, enhancing the user experience and enabling effective product search